

Video Slides W5: Surplus

ECO 101: Principles of Microeconomics

Prof. Freitas, University of Toronto

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Surplus Definitions

Topic: Surplus

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Video Learning Objectives

- Define Surplus, Consumer Surplus, Producer Surplus and Total Surplus.
- Calculate and graphically identify them for individual consumers, firms and in the market.

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Why Surplus?

- Market: voluntary trades that lead to equilibrium prices.

Questions

- Are these trades making people better off?
- Can we quantify how much better off?

Surplus

- Economists use surplus as
 - a way to measure who benefits from a transaction and by how much.
 - compare allocations.
- Why not use price as a way to measure benefits?.

Price vs. Surplus

Example: Home internet

- Price: Home internet cost average \$95 a month.
- Your surplus:

Individual Surplus

Consumer Surplus (CS)

- From a given unit: $MWTP - P$
- Total CS: $WTP - \text{total price paid}$

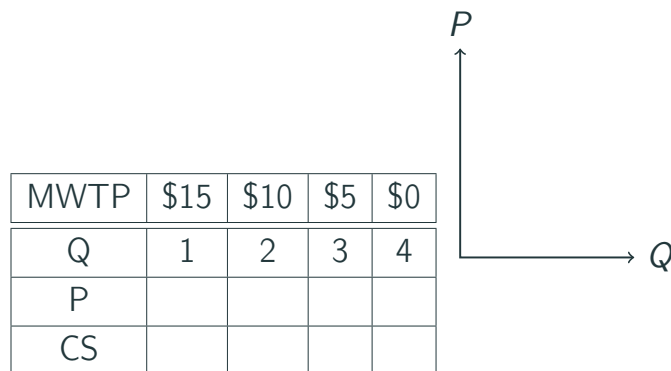
Producer Surplus (PS)

- From a given unit: $P - MC$
- Total PS: Revenue $-$ Area under the MC curve

Consumer Surplus

MWTP: Marginal willingness to pay

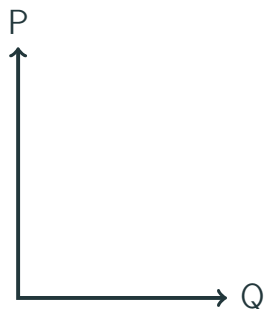
Discrete Data



Consumer Surplus

MWTP: Marginal Willingness to Pay

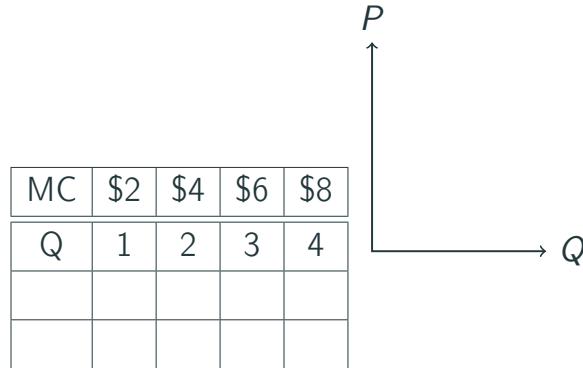
Continuous data $MWTP(Q) = P(Q)$



Producer Surplus

MC: Marginal Cost

Discrete Data



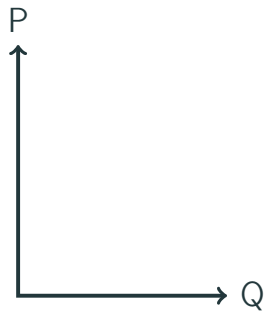
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Graph: Producer Surplus

MC: Marginal Cost

Continuous data $MC(Q) = P(Q)$



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Societal (Total) Surplus

- Societal Surplus is also called Total Surplus
- It includes the surplus every single member of society gets due to the trade, regardless of whether they participate in the trade or not

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Societal (Total) Surplus

- From a given unit: Social Marginal Benefit — Social Marginal Cost
- Total Surplus: Total Social Benefit - Total Social Cost

Societal (Total) Surplus

MSC: Marginal Social Cost, MSB: Marginal Social Benefit,
MWTP: Marginal willingness to pay, MC: Marginal Cost

Questions

- What is the relationship between *MSC* and *MC* (similarly *MWTP* and *MSB*)?
- Isn't Total Surplus = Consumer Surplus and Producer Surplus?

Summary

MSC: Marginal Social Cost, MSB: Marginal Social Benefit,
MWTP: Marginal willingness to pay, MC: Marginal Cost

- To calculate Consumer Surplus and Producer Surplus we need to know prices, MWTP and MC.
- To calculate Total Surplus we need quantity and how all members of society are affected by the transaction i.e. MSB and MSC.

Efficiency

Topic: Surplus

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Video Learning Objectives

- Calculate the change in surplus due to a change in allocations.
- Define efficiency in terms of surplus.
- Identify efficient/inefficient allocations.

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Individual Trades

MWTP: Marginal Willingness to Pay, MC: Marginal Cost, CS:

Consumer Surplus, PS: Producer Surplus

Suppose two people meet and decide to trade voluntarily.
How does surplus change when this happens?

- “Consumer”
- “Producer”
- Trade voluntarily:
- Price?
- “Total Surplus”

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Individual Trades

MWTP: Marginal Willingness to Pay, MC: Marginal Cost, CS:

Consumer Surplus, PS: Producer Surplus

Suppose we take a good from A and give it to B. Neither has a say in the matter. How does surplus change when this happens?

- “Consumer”
- “Producer”
- Price?
- “Total Surplus”

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Price and Surplus

- The Total Surplus from a given trade is unaffected by price.
- Prices only serve to allocate surplus between consumers and producers

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Allocation Questions

Two sides of a market (e.g. consumers and producers)

- Given the unit is produced, who should get it ?
- If the unit has to be produced, who should produce it?
- Should the unit be produced?

Short version: Is the allocation “good”?

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Allocating Goods Already Produced

Example:

- Good: 1 ticket for a Zaz concert.
- Manon: has ticket paid 50\$, values concert at \$70.
- Me: No ticket, values concert at \$150.

Q: Who should get the ticket?

Who should produce the good?

- Unit must be produced, who should produce it?

Example:

- Good: 1 hour of tutoring, student pays \$80 per hour
- Janina: Marginal (Opp.) Cost = \$30
- Gosia: Marginal (Opp.) Cost = \$50

Reallocations

MSC: Marginal Social Cost, MSB: Marginal Social Benefit,
MWTP: Marginal Willingness to Pay, MC: Marginal Cost

Reallocating goods

- Already produced, to someone with a higher *MWTP* increases total surplus.
- To be produced from a high *MC* producer to a low *MC* producer increases total surplus.

Individual buyers and producers may not like reallocation depending on how it is done.

Allocation Questions

Two sides of a market (e.g. consumers and producers)

- Given the unit is produced, who should get it ?
- If the unit has to be produced, who should produce it?
- Should the unit be produced?

Short version: Is the allocation “good”?

Should an Extra Unit be Produced?

MWTP: Marginal Willingness to Pay, MC: Marginal Cost

Start with a single transaction. **B**en wants the good, **S**ue has a good.

- **B**en's benefit i.e. marginal private benefit = $MWTP$
- **S**ue's cost i.e. marginal private cost = MC
- Transaction will take place if they can find a price such that $MWTP \geq p \geq MC$.
- Should the transaction take place?

Should an Extra Unit be Produced?

MSC: Marginal Social Cost, MSB: Marginal Social Benefit,

MWTP: Marginal Willingness to Pay, MC: Marginal Cost

(**Private**) **surplus** from the transaction: $MWTP - MC$

(**Total**) **Societal surplus** generated by exchange:
 $MSB - MSC$

- Society wide: Transaction increases surplus if $MSB > MSC$

Should an Extra Unit be Produced?

MSC: Marginal Social Cost, MSB: Marginal Social Benefit

- If a unit has $MSB > MSC$, producing it increases Total Surplus.

What is the relationship between private and social benefit?

Assumption - No Externalities

MWTP: Marginal Willingness to Pay, MC: Marginal Cost

- Ben (and only Ben) gets all the benefits from the transferred good.
 - Marginal Social Benefit = Marginal Private Benefit (= $MWTP$)
- All the costs of producing and giving up the good are borne by Sue (and only Sue)
 - Marginal Social Cost = Marginal Private Cost (= MC)
- This is not the case for many cases, e.g. pollution from gasoline

Assumption - No Externalities

MSC: Marginal Social Cost, MSB: Marginal Social Benefit,

MWTP: Marginal Willingness to Pay, MC: Marginal Cost

- We will (for now) make the perfectly competitive market assumption - no external costs and benefits from a transaction.
- Assumption $\Rightarrow MSB = MWTP$ and $MSC = MC$.

Is the Given Allocation “good”?

Two sides of a market (e.g. consumers and producers)

- “Good” from the perspective of society
- Efficiency as **one** measure of a “goodness”. Uses surplus.
- BUT: **Efficient does not mean “fair”**.

Efficiency

MSC: Marginal Social Cost, MSB: Marginal Social Benefit

An allocation is efficient if

- It maximizes (Societal) Total Surplus $\Rightarrow$ produce and trade all goods where $MSB \geq MSC$ and not any others
- No way to make at least one person better off without making anyone worse off.

Are Markets Efficient?

“Good” for society

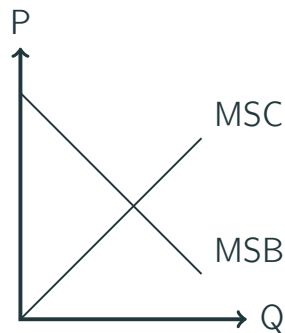
- Claim: a **perfectly competitive market** allocation maximizes efficiency.
- BUT: based on many assumptions.

Efficient Quantity

MSC: Marginal Social Cost, MSB: Marginal Social Benefit,

TS: Total Surplus

What is the efficient quantity?



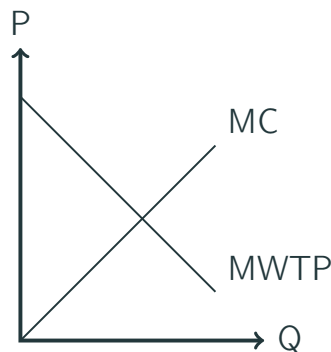
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Market Quantity

MWTP: Marginal Willingness to Pay, MC: Marginal Cost

What is the market quantity?



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Efficiency-Thought Experiment

MSC: Marginal Social Cost, MSB: Marginal Social Benefit,

TS: Total Surplus

- If **no externalities**
 $MWTP = MSB, MSC = MC \Rightarrow Q^* = Q^{eff}$
- Markets produce the efficient quantity, Total Surplus is maximized.
- **NOTE:** Consumer and Producer Surplus are not maximized at the efficient quantity

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Summary

- We can increase efficiency by producing good where $MSB \geq MSC$
- Under the competitive market **assumptions**, market allocation maximize total surplus.
- Price does not determine surplus, but allocates surplus from a transaction between consumers and producers.

Market Failure

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Video Learning Objectives

- Discuss: do markets always lead to efficient outcomes?
- Explain why and how to use deadweight loss as a measure of inefficiency.
- Discuss: Efficiency and “Fairness”

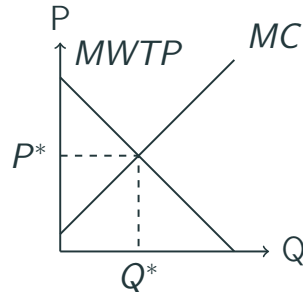
Markets and Efficiency

MWTP: Marginal Willingness to Pay, MC: Marginal Cost, MSC:

Marginal Social Cost, MSB: Marginal Social Benefit

No externalities $\Rightarrow$

$$MWTP = MSB, MSC = MC \Rightarrow Q^* = Q^{eff}$$



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Market Failure

- Market failure occurs when the market results in the inefficient quantity.
- Typically this happens when the assumptions of Perfect Competition do not hold.

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Market Failure

Examples of causes that lead to market failure.

- Market power.
- Externalities.
- Informational problems that undermine trust.
- Government regulation.

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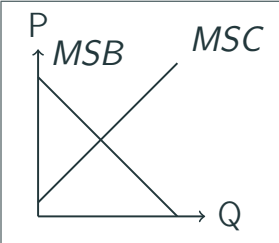
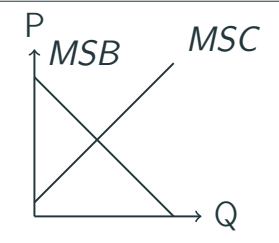
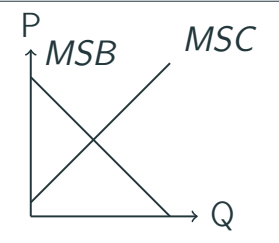
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Measuring Inefficiency

- Deadweight Loss: Loss in surplus relative to the efficient surplus.
 - Trades for which $MSB > MSC$ do not happen or those for which $MSB < MSC$ take place.
 - The wrong people get/produce the good.
- Notice everything points to quantity not price.

Efficiency

MSC: Marginal Social Cost, MSB: Marginal Social Benefit

$Q < Q_{\text{efficient}}$	$Q > Q_{\text{efficient}}$	Misallocation
		

Is Efficiency Everything?

Aren't some things more important than efficiency?

Efficiency and “Fairness”

MWTP: Marginal Willingness to Pay

- Market allocation result in people with the highest *MWTP* getting the good.
- What determines *MWTP*?
 - likes and dislikes (preferences) holding fixed ability to pay
 - ability to pay (i.e income) holding fixed preferences

Efficiency and “Fairness”

MWTP: Marginal Willingness to Pay

- Ability to pay could be determined by choices, but also by luck...(e.g. the country, family you're born into)
- What about other measures of “deserving”?

Economists and “Fairness”

- Economists focus on efficiency leaving “fairness” for society to define.
- Given what society “wants” economists can help find the most efficient way to get there.

Summary

- Without all the competitive assumptions holding, markets won't necessarily be efficient.
- Efficiency $\neq$ Fairness